

WELL COMPLETION REPORT (SYNTHETIC)

OIL-NHK-011 - Nahorkatiya-011

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000122
Field / District	Nahorkatiya / Dibrugarh
Block	PEL-ASSAM-02
Basin	Assam-Arakan
Formation	Girujan (GRJ)
Spud / Completion	2021-06-03 / 2021-08-10
Measured / TVD	2369 m / 2324 m
Trajectory	Directional; inc 20.9 deg; az 31.9 deg
Formation Interval	1120-1978 m
Lithology	Silty shale with sandstone lenses
Pore Pressure Reference	2285 psi
Fracture Gradient	15.28 ppg
Mud System	KCl Polymer WBM
NWIS Well Risk	Moderate (47.0/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	592	Surface
9 5/8 in casing	1374	Intermediate
7 in liner/casing	2084	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2021-07-16 1581 m	Mud Loss	High	Flow-out dropped below flow-in with pit loss. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Track flow balance and pit volume trend before losses become severe.
2021-07-21 1737 m	Mud Loss	High	Flow-out dropped below flow-in with pit loss. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Track flow balance and pit volume trend before losses become severe.
2021-07-28 1961 m	Stuck Pipe	High	High drag was recorded while circulating remained possible. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency.
2021-07-29 1996 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Track flow balance and pit volume trend before losses become severe.
2021-06-11 406 m	Fishing Operation	Medium	Downhole tool remained unrecovered after freeing attempts. Mitigation: Maintain clear event chronology and NPT accounting during recovery. Lesson: Capture retrieval outcome as institutional knowledge.
2021-06-24 864 m	Torque Spike	Medium	Rotary speed fluctuated during constant pump rate. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Use offset torque trend to set adaptive operating envelope.
2021-07-15 1546 m	Torque Spike	Medium	Rotary speed fluctuated during constant pump rate. Mitigation: Reduce WOB/RPM and circulate before resuming normal parameters. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.

Completion Summary

The synthetic well reached 2369 m MD in the Girujan interval. The NWIS knowledge record links 12 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.